NYU TANDON SCHOOL OF ENGINEERING

NYU TANDON CAREER HUB · FALL 2026

Data Science Bootcamp

Week 3 · Exploratory Data Analysis & Data Visualization

NASA Exoplanet Analysis

Instructor: **Angelin Ann Jacob**

Section 1 · Tue Oct 13

Section 2 · Thu Oct 15

Useful Links

Zoom
All sessions are virtual. Use ONLY **your NYU SSO Login to be considered for attendance**

Brightspace
Assignments, submissions and the discussion board, monitored Monday to Thursday.

GitHub
Slides and Colab notebooks for every session.

Email
datasciencebootcamp@nyu.edu for questions, extensions and attendance approvals.

1:1 coaching
Optional 30-minute technical interview prep. Book it through Handshake.

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Exploratory Data Analysis

Get to know a dataset before you trust it, model it or present it!

Data Science Bootcamp · Fall 2026

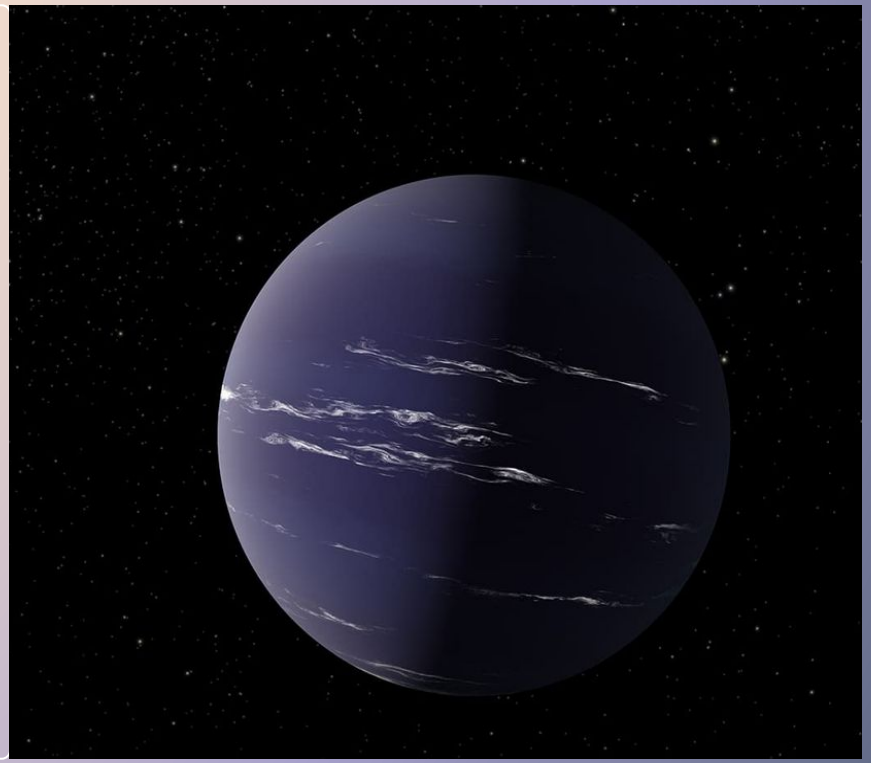
NASA Exoplanet Archive

https://exoplanetarchive.ipac.caltech.edu

Dataset name: Confirmed planets, one row per planet (exoplanetdata, ModernDive 2026 snapshot)

Data Provided By:
NASA Exoplanet Science Institute (Caltech / IPAC)

Size: 6,278 planets, 28 columns, 192 of them found in 2026
Visit: github.com/moderndive/exoplanetdata



HOW TO READ TODAY'S SESSION

Predict, code, share

PREDICT

Guess before you run

Post your guess in the Zoom chat, then run the cell and compare.
How many planets are there? Which method misses the most data?

YOUR TURN

Write the code, then share

Post your answer in the Zoom chat.

Hint open it if you are stuck
Answer open it only to check

EXAMPLE

Story: What share of planets were photographed?

```
(planets["discovery_method"]
    == "Imaging").mean()
```

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Today's plan: the EDA workflow

2 · The dataset

What did we load? Rows, columns, data types and summary statistics.

3 · Data quality

Can we trust it? Data Story 1: missing values and duplicates.

4 · One column

What does one column look like? Histograms, log scales, box plots.

5 · Two columns

How do two columns move together? Data Story 2: scatter plots, correlation.

6 · Insights

What did we learn? Short insights, each backed by a number.

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SECTION 2 · UNDERSTANDING THE DATASET

Meet the planets

THE STORY	THE TOOL
What did we just load? How big is it?	`planets_raw.head()` `planets_raw.sample(5)` `planets_raw.shape` `planets_raw.dtypes`
Why is discovery_year a decimal (2007.0)?	`planets["discovery_year"].astype("Int64")`
Typical values. Is the mean far from the median?	`planets.describe().T` `# period: mean 72,082 days, median 10.7`
What does one planet's card look like?	`PLANET = "51 Peg b"` `planets.loc[planets["planet_name"] == PLANET].T`

SECTION 3 · DATA QUALITY · DATA STORY 1

Most exoplanets have never been seen?

Transit	Radian velocity	Microlens	Imaging
5%	83%	98%	35%
0.08 AU	0.9 AU	2.3 AU	156 AU

THE STORY

Are the gaps random, or tied to the method?

THE TOOL

```
main.groupby(
  "discovery_method")
  ["eq_temp_k"].apply(...)
```

THE STORY

Can a value be missing and still look real?

THE TOOL

```
DISCOVERY_METHOD =
"Microlensing"
sizes.nunique()
```

SECTION 4 · ONE VARIABLE AT A TIME

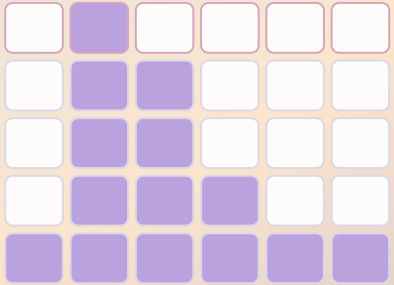
Most planets finish a year in days

0.09	4.3	10.7	38	…	402M

Orbital period in days: min, Q1, median, Q3 … max

`np.log10(period)`

under 1 day → over 10,000 days



taller column = more planets

normal scale

one 402-million-day year squashes it all into one bar

log scale

one clear peak: a few days per year

THE STORY

How long is a typical exoplanet year?

Median: 10.7 days. 83% of planets have a shorter year than Mercury (88 days).

THE TOOL

```
period.median()              # 10.7
(period < 88).mean()         # 0.83
plt.hist(np.log10(period), bins=50)
sns.boxplot(data=main, x=..., y=...)
```

SECTION 5 · TWO VARIABLES AT A TIME · DATA STORY 2

Hot Jupiters: the planets nobody expected

Every planet

one row = one planet

→

```
planets["hot_jupiter"]
```

radius > 8 Earths, year < 10 days

→

```
len(hj)   # 739
```

hot Jupiters in the data

THE STORY

Do planets with shorter years run hotter?

THE TOOL

```
plt.scatter(period, temp)
ax.set(xscale="log")
# r = -0.68
```

THE STORY

How hot is a typical hot Jupiter?

THE TOOL

```
hj["eq_temp_k"].median()
# 1,460 K
.nlargest(1, "eq_temp_k")
```

THE STORY

Which columns carry the same information?

THE TOOL

```
sns.heatmap(logs.corr(),
            annot=True)
# period vs orbit: 0.99
```

SECTION 6 · DRAWING INSIGHTS

What we learned today

THE INSIGHT	THE EVIDENCE
Most exoplanets have never been seen	only **1.5%** were photographed (imaging)
Missing values depend on the discovery method	temperature missing: **5%** of transit, **98%** of microlensing planets
Most known planets finish a year in days	median year **10.7** days; **83%** shorter than Mercury's 88-day year
Closer orbits mean hotter planets	r = **-0.68**; **739** hot Jupiters with a median of **1,460 K** and a **3.5**-day year

Before next week

Take-home

Exercises 1 to 4 in today's notebook. Submit on Brightspace within one week, by 11:59 PM.

Feedback form

Fill it out after today's session. It counts toward your credential.

Week 4: SQL Fundamentals

Work with databases and get familiar with SQL basics!